POLITICAL PHILOSOPHY · INSTITUTIONAL ECONOMICS

**The new social contract
in the age of artificial
intelligence**

An analysis of the clauses of a
prosperous, free, and enduring civilization

21 clauses
Rights, obligations, and the distribution of abundance
Control of power and the continuity of civilization

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Introduction. The purpose and method of the proposal

A new social contract becomes necessary not when a spectacular technology appears, but when the relationship among contribution, dependence, and power changes. If a substantial share of goods and services can be produced with much less human labor, wages can no longer remain the only legitimate bridge between a person and the means of a decent life. At the same time, whoever controls automated infrastructure may acquire more power than a traditional employer: they may simultaneously control production, distribution, information, and the environment in which people form their preferences. This book examines the conditions under which this transformation could support a prosperous and stable civilization without turning most of the population into dependent beneficiaries of a leadership that cannot be replaced.

The working hypothesis is favorable: artificial intelligence becomes highly productive, and automation makes it possible to meet many needs at lower real costs. This is a design assumption, not a dated prediction. Empirical findings about AI assistants in bounded activities show the possibility of productivity gains, with differing effects among workers; they do not demonstrate the disappearance of work or general material abundance. The study by Brynjolfsson, Li, and Raymond on customer support is relevant precisely as local evidence, not as a certificate for a future economy. [3]

The abundance examined here concerns capabilities, not infinity. Electricity, land, water, materials, ecosystems, and human time do not become unlimited because producing a

digital response becomes cheaper. The IEA's 2025 report on energy and AI emphasizes digital infrastructure's dependence on electricity and grid capacity. This book does not treat any energy trajectory as destiny; it uses this dependence to rule out the false premise that automation would suspend physical constraints. [5]

From controlling information to controlling the willingness to judge

The political starting point is the idea that information can be neutralized without being banned. A true message may remain accessible yet become practically unnoticeable in an environment dominated by competing feeds, emotional demands, and personalized entertainment. Three phenomena must nevertheless be distinguished: censorship proper, strategic influence over attention, and crowding produced without coordination by commercial competition. Calling all three censorship would confuse intention, mechanism, and effect and make the diagnosis impossible to test.

There is evidence of deliberate use of distraction. King, Pan, and Roberts analyzed an operation producing social media messages in China and identified a strategy aimed at changing the subject and symbolic mobilization rather than arguing with critics. The finding supports the existence of a mechanism of control through distraction in a defined context; it does not show that all platforms or societies work this way. Extending the conclusion to a future system remains an institutional hypothesis that must be examined separately. [1]

Nor will the phrase 'cheap dopamine' be used as a sufficient scientific explanation. It intuitively expresses easy access to repeated rewards, but does not by itself identify the cause of a behavior, the degree of autonomy, or the social effect. A serious analysis examines the architecture of choice, the provider's incentives, the user's ability to stop, and whether their behavior agrees with the preferences they can endorse after reflection. A long period of use does not automatically prove either well-being or addiction.

An experiment in which Facebook was deactivated, published by Allcott and colleagues in 2020, found both increased subjective well-being and reduced news knowledge. That combination matters: leaving an information environment can help in one respect and harm in another. It does not yield a universal recommendation to withdraw, but a need to evaluate information, subjective experience, and practical freedom separately. [2]

The political offer to be evaluated

The implicit offer of an automated order might be expressed as follows: the population delegates governance to highly competent institutions, and they ensure material comfort, safety, and access to varied experiences. People can devote their time to relationships, communities, virtual worlds, games, research, or personal projects with no economic return. The offer should not be rejected merely because it includes comfort and entertainment. Reducing the obligation to work for survival is a serious goal; the obligation to manage society continuously must not become a new form of work imposed on everyone.

The problem is what delegation transfers. The administration of services can be delegated. The final determination of collective goals, the removal of the possibility of opposition, and making material existence conditional on loyalty are not mere extensions of the same operation. A limited mandate can spare the public's attention. An irrevocable transfer of authority can eliminate the very means by which the population might discover that the original offer is no longer being honored.

The book's thesis is that broad delegation can be justified only if it remains accompanied by effective rights of oversight, contestation, replacement, and exit. The right not to take part in politics every day is compatible with retaining ultimate political authority. Temporarily not exercising a right must not be interpreted as surrendering it. This is a normative position defended in the book, not an inevitable result of AI development.

What a rigorous clause means here

The social contract is not treated as a commercial contract that every person has signed. Newborns, dependent people, and future generations cannot participate on equal terms in such a negotiation. The term denotes a justifiable order of rights, obligations, and public powers. Its acceptability is assessed both from the perspective of those who can negotiate and from that of those whose interests must be represented without present consent.

Each clause is examined using the same logic: the problem it solves; its principle of justification; the holder of the right; the institution obliged to respect it; the material conditions for implementation; the main objection; and the mechanism for verification and remedy. In the body of the book, these dimensions are developed as arguments rather than reduced to administrative forms. At the end, the clauses are brought together in a compact normative proposal. Definitive legal wording would require constitutional and legislative adaptation in jurisdictions that decided to adopt it.

Three levels are explicitly separated. Empirical data and references to legal instruments are supported by sources. Numerical models are analytical exercises with their assumptions stated, not forecasts. The clauses are normative proposals. The existence of an institutional precedent does not prove it will work on a planetary scale; the desirability of a right does not prove it can be provided instantly at every level of consumption.

A horizon of thousands or millions of years is used to test the cumulative consequences of choices, not to claim predictive ability over such intervals. Stability will mean continuity of life, cooperation, and the possibility of correction. It will not mean preserving the same leadership, culture, or definition of a good life. An order that survives only by prohibiting change can be administratively durable and still incompatible with this contract's purpose.

Chapter I. The status of the person and the limits of delegation

The first problem is not how many goods can be distributed, but the status by virtue of which they are distributed. A benefit offered out of generosity, a recognized right, and a service granted in exchange for obedience may have the same material appearance. They produce different relations of power, however. The three clauses of this chapter define a person as a holder of rights, establish the scope of the political mandate, and require that refusal of an institution not become equivalent to exclusion from society.

Clause 1. Equal status, independent of economic usefulness

No person loses civic status, legal protection, or access to the essential conditions of existence because their labor is no longer demanded economically, because they have reduced capabilities, or because they reject a life project approved by the authorities.

Why productivity cannot be the basis of status

If rights were proportional to measurable economic contribution, automation could turn a technical improvement into political degradation for those displaced. People would become less entitled to exist precisely when society could sustain their existence more easily. This conclusion is not a logical consequence of efficiency, but the result of a distributive rule that confuses the price of a service with the value of the person who provides it.

The argument does not assume that all contributions are identical or that every difference in remuneration is unjust. It separates status from contractual reward. Someone may

receive more for difficult work without receiving more fundamental rights. Someone who does not perform such work may have a limited duty to contribute, but does not thereby become an object available for social optimization. Basic protection precedes the calculation of remuneration.

A general legal reference for this distinction exists in the principles of equal dignity, recognition before the law, and social protection in the Universal Declaration of Human Rights. The present proposal extends the analysis of these principles to dependence created by automated infrastructure; it does not claim that the declaration already contains the entire regime described here. [6]

An additional justification is impartiality toward the position we might occupy. We need not accept Rawls's theory in its entirety to use the question it makes central: what rules would be justifiable without relying on the advantage of knowing we would occupy a favorable position? In an automated economy, technical advantage, health, and property are not conditions any individual can guarantee throughout a lifetime. [9]

Equality of status and justified differentiation

The clause does not require the same material provision for everyone. Strict accounting equality can produce real inequality if some people need adaptations, care, or additional assistance to use a service. The proposed criterion is effective access to essential functions, not the mechanical distribution of the same amount. Differential treatment must be justified by verifiable needs and obstacles, not by a moral judgment about the person.

Children enter the contract as holders of their own interests, not merely as extensions of their parents. People unable to express full consent need representation, protection against conflicts of interest, and adapted ways of being heard. Inability to negotiate cannot be a reason for exclusion: it is precisely what justifies an additional duty of protection. An order that defended only participants capable of threatening to withdraw would not be a social contract but an agreement among powerful actors.

Equal status must also apply to those who criticize the system or choose unpopular ways of life. Denying basic housing because someone advocates a lawful doctrine, publishes criticism, or prefers to live in seclusion would turn social policy into a disciplinary instrument. Legitimate limits concern conduct that harms others, not a lack of enthusiasm for the existing order.

The obligated institution and the test for violation

The primary obligation falls on the public authority that determines access to services, but the clause must also apply to delegated private operators. Outsourcing delivery cannot outsource constitutional accountability. If a provider manages the sole identity or distribution infrastructure, its commercial contract must be subordinate to safeguards protecting people from arbitrary exclusion.

The test for violation is concrete: do two people with relevantly similar needs receive different treatment because of their opinions, membership, productivity, or a general compliance score? An individual error must be correctable quickly, and a discriminatory rule must be suspendable. An audit must also examine people who never succeeded in filing an application, not only completed cases. Otherwise, administrative exclusion vanishes from the statistics.

The strongest objection is that unconditional material status would weaken reciprocity. The answer is not to deny costs but to separate the basic right from proportionate duties and additional rewards. Those duties are examined in Clause 6. No rule of reciprocity should, however, permit hunger or abandonment of medical care to become ordinary means of securing compliance.

Clause 2. Delegating administration, not surrendering authority

Administrative powers may be delegated to human institutions, AI systems, or mixed structures only through bounded, verifiable, and revocable mandates. Material benefits cannot be made conditional on relinquishing political control over those mandates.

What can justify very broad delegation

A complex society cannot ask every resident to approve every operational decision. Energy distribution, infrastructure maintenance, and service coordination require continuity and specialization. AI may expand the range of delegable tasks, especially when results are measurable and errors can be detected. It is reasonable for people to prefer these operations to work without their constant personal involvement.

Nevertheless, three levels of authority must be distinguished. At the operational level, means are chosen within a mandate. At the public-policy level, priorities are set among competing goals. At the constitutional level, it is decided who has the right to set those priorities and what even a majority cannot decide. Performance at the first level does not automatically confer legitimacy at the other two. A system can estimate the costs of a policy excellently without having the authority to decide who should bear them.

The offer 'let us govern and we will ensure you a good life' therefore has an acceptable version and a problematic one. The acceptable version is a broad mandate to deliver services accompanied by independent oversight. The problematic version is an exchange in which present well-being buys the impossibility of future revocation. The difference is not the degree of comfort but the distribution of decision rights in case of conflict.

The right not to participate continually

It does not follow that everyone must become an amateur administrator. It would be contradictory to celebrate the time freed by automation and then impose a permanent duty to deliberate. The contract must protect intermittent political participation: the possibility of delegating, following only certain issues, and returning when reasons for concern arise. The right to participate in public affairs offers a reference point, but the institutional form proposed here is more specific than existing international instruments. [7]

The obvious risk is that formal oversight decays when too few people follow those in charge. Therefore, freedom from individual participation must be offset by institutions funded to provide scrutiny: an independent press, public audit, critical research, opposition, and deliberative bodies. These are not an elite authorized to replace the population, but infrastructure that reduces the cost of returning to the decision-making process.

OECD research on representative deliberative processes documents ways in which groups of citizens can examine complex issues with adequate time and support. This precedent does not show that such groups should replace elections or judicial review. It does, however, suggest that informed participation can be organized rather than assumed to arise from everyone's spontaneous availability. [11]

The mandate and conditions for revocation

A serious mandate specifies its objective, resources, limits, evaluation criteria, and the authority that may intervene. 'Maximize well-being' is insufficient: it does not say whether the system may sacrifice a minority, change the population's preferences, or conceal information to obtain an apparently better outcome. A mandate must contain explicit prohibitions and duties to report uncertainty, not merely a performance function.

Revocability must be technical and financial, not just legal. An operator impossible to replace because of proprietary formats, exclusive control of access keys, or the absence of alternative personnel may hold real power even if its mandate formally expires. For that reason, periodic transfers of operation must be practiced, and contracts must provide an authorized successor access to what continuity requires.

The objection about instability is legitimate: an administration removed for every dissatisfaction cannot plan. The proposed solution is not permanence but separating time horizons. Current operations may have contractual stability; distributive policies may be reviewed periodically; serious rights violations may trigger immediate intervention. The frequency of oversight must reflect the consequences, not the technical capacity to hold endless consultations.

The decisive indicator is a real capacity to replace leadership without losing basic services. An order that provides comfort only while the same center remains untouched does not meet this clause, however satisfactory its average outcomes may be.

Clause 3. Effective exit and a plurality of alternatives

A person must be able to leave a provider, platform, or community without unjustifiably losing access to rights, identity, their own property, and transferable relationships. The right to exit does not include a right to avoid legitimate obligations toward others.

Why a nominal alternative is insufficient

Freedom of choice can exist on paper and disappear in practice. Two services may have different names yet depend on the same identity infrastructure, the same AI provider, and the same suspension mechanism. In that case, changing the interface does not reduce dependence. The contract must examine who can actually refuse service and who can prevent the use of an alternative.

Exit involves several costs: learning, lost data, disrupted relationships, abandoned digital property, and the risk of losing access to essential services. Some costs are unavoidable; leaving a community cannot preserve every relationship intact. Others are deliberately manufactured to block departure. The relevant distinction is between the inherent difficulty of change and an obstacle the provider could remove without violating others' rights.

The proposal does not promise full portability of one's life. A shared conversation may contain information belonging to other participants, and transferring it must respect their interests too. A work created in a virtual environment may depend on separate licenses. Social relationships are not objects owned exclusively by the user. For precisely this reason, portability must be designed by category: personal data, one's own property, identity, authorizations, and social links for which those involved have agreed.

The infrastructure of alternatives

A minimal public alternative is justified where exclusion from the market would prevent the exercise of rights. It need not be the most appealing service, but it must permit essential communication, authentication, and access to lawful benefits. Funding it is the cost of effective freedom, conceptually comparable to maintaining a court to which someone may turn even if most disputes are settled privately.

At the same time, open standards and separation of components can reduce dependence on a single operator. They are not miraculous solutions. A standard may be informally controlled by dominant actors, and open software may depend on inaccessible physical infrastructure. An audit of alternatives must include operating capacity, not merely code availability or the formal existence of several licenses.

In local communities, exit also requires material rights. If housing, income, and care depend on community leadership, freedom to leave can be purely theoretical. Transition support, the possibility of renting a home elsewhere, and access to an external jurisdiction reduce this dependence. The requirement is not unlimited funding for every move but prevention of a situation in which a person's survival depends on accepting local domination.

Limits and verification

The right to exit cannot extinguish parental duties, legally established debts, or liability for harm. Nor does leaving a community confer a right to destroy shared infrastructure or transfer other people's secrets. The contract must make exit possible without turning it into a means of shifting costs onto those who remain.

The economic objection concerns the provider's investments: if users can leave easily, who funds development? The answer is that payment should rest on the service's value and explicit contractual terms, not on the user's inability to abandon it. Specific investments may justify reasonable periods or transparent termination fees; they do not justify monopolizing identity and fundamental rights.

The clause is tested through migration exercises with diverse users and measurement of the actual costs of leaving. Successful transfers are not the only cases that must be documented; reasons for giving up must be documented too. A system that reports high satisfaction after dissatisfied people have been prevented from leaving does not provide evidence of consent. It provides, at most, a statistic about the population remaining inside it.

Chapter II. Material security and ownership of abundance

The distribution of abundance cannot be treated as a secondary stage that automatically follows production. The technical capacity to produce, the right to use a product, and the power to set the conditions of its use are distinct. An economy can become highly productive yet maintain severe social scarcity through ownership or access rules. The contract must establish both what is guaranteed to the person and the mechanism through which society participates in the revenues of the infrastructure that makes that guarantee possible.

Clause 4. A guaranteed and adaptable material floor

Every person under the contract's jurisdiction has effective access to an essential package of goods and services, publicly defined and sustainably funded. Its level is updated according to real delivery capacity, needs, and ecological constraints, without political conditions.

What is guaranteed: functions, not unlimited consumption

A material guarantee should be formulated in terms of the functions it enables: adequate food, shelter, water, essential energy, care, education, communication, and access to justice. Defining it by functions avoids two errors. The first is promising a nominal sum that loses value when prices rise. The second is imposing a single mode of delivery regardless of preferences and local conditions. The package may combine cash benefits, services, and access to infrastructure.

The right to housing does not mean a right for everyone to the same address or an arbitrarily large area. Access to health care does not amount to a right to every imaginable intervention regardless of evidence and cost. A rigorous formulation distinguishes an enforceable core, progressive expansion, and additional choices a person funds from other income. Without this separation, each person's right becomes incompatible with the same right for everyone.

The International Covenant on Economic, Social and Cultural Rights provides a reference point through progressive realization and the use of available resources. This book proposes a more explicit way to operationalize the relationship between a right and delivery capacity. It neither reduces all existing obligations to mere budget targets nor claims that citing a lack of funds justifies every regression. [8]

Services, income, or a combination

A basic income offers freedom of choice and may reduce administrative control over personal life. But where supply is rigid, a cash transfer may be absorbed by prices or rents. Well-located homes do not automatically multiply when disposable income rises. In such a sector, the guarantee also requires investment in supply, access to land, infrastructure, and competition rules. Payment cannot substitute for production.

Universal services can protect access in essential sectors and distribute risk among beneficiaries. They can, however, become bureaucratic, paternalistic, or insensitive to preferences. Direct delivery must be accompanied by quality standards, alternatives, and a right to contest denial. The model preferred here is mixed: services for major risks and essential infrastructure, cash for everyday choices, and justified supplements for special needs.

The Finnish basic-income experiment found limited employment effects and favorable results for perceived well-being. The official evaluation, however, describes a bounded program within an existing social system, not an economy in which automation has replaced wages. The result supports institutional experimentation, not the conclusion that any universal income at any level can be funded without side effects. [12]

The financial test and the material test

Feasibility requires two independent checks. The first concerns recurring financial resources. The second concerns the capacity to deliver the goods and services. A balanced budget does not guarantee enough caregivers, available energy, or adequate housing. Conversely, the existence of goods does not guarantee legitimate distribution if access is legally or economically blocked. Both sets of accounts must be published together.

A simplified accounting model can clarify the first check. Let T denote recurring tax revenue, F a sustainable transfer from the common fund, S the cost of essential services, I public investment, R reserves, and H debt service. If N is the eligible population, the annual cash dividend per person b must satisfy the following relationship for a budget with no new debt financing:

$$N \times b + S + I + R + H \leq T + F$$

The relationship is not a complete macroeconomic theory. It does not explicitly include foreign trade, price responses, or behavioral changes; these must be analyzed separately. Its use is to prevent incompatible uses of the same revenue from being promised simultaneously. If a tax is calculated before a public fund's profits are distributed, the same income cannot be counted in full again as the fund's dividend.

Consider a purely illustrative example expressed in billions of monetary units per year. T is 300, F is 30, S is 140, I is 70, R is 20, and H is 20. That leaves 80 for cash transfers. For ten million beneficiaries, b is 8,000 per year, or about 667 per month. The sum is neither a recommendation nor a description of a country; it shows only how the guarantee is tied to a fundable base.

If revenues T and F fall by 20%, total resources become 264. Keeping the other uses unchanged leaves 14 for the dividend, not 80. This result shows why a stable promise cannot rest solely on volatile revenues. The system would need accumulated reserves, stabilization rules, rescheduling of some investments, and possibly temporary debt financing compatible with solvency. None of these options automatically creates missing goods.

Updating and remedy

The package is updated through a public procedure based on the cost of services, technological progress, and needs. A temporary reduction must be justified with verifiable information and the alternatives considered. The proposed priority is continuity of essential functions, not preserving the same cash sum at any cost. Adjustment must not be left to an algorithm that automatically turns a fall in revenue into a loss of care.

Relevant indicators are actual shortages, access time, and quality differences among groups. A right can be violated even if the entire budget is spent. A person must be able to obtain provisional support when delay causes serious harm, and the administration must bear the duty to explain a denial. The material guarantee thus becomes a verifiable obligation, not a formula of legitimacy without consequences.

Clause 5. A stake in common productive capital

A share of the revenue generated by productive capital, common resources, and public investment must support widely distributed benefits through transparent ownership, taxation, or participation mechanisms. Neither public nor private ownership confers a right to dominate access to the essential conditions of life.

Why transfers alone are insufficient

If the population's income depends less and less on wages, its source must shift toward other claims on the value produced. Fiscal transfers are one option, but not the only one. Stakes in capital, dividends from collective funds, cooperatives, and shared access to infrastructure can distribute benefits before or alongside budget redistribution. The choice

affects power: someone receiving an annual payment does not necessarily influence the rules that determine it.

The IMF's 2024 fiscal analysis of AI discusses social protection and the distribution of gains, including issues concerning capital and taxation. It does not provide a single model of an abundance economy. Its relevance to the contract lies in recognizing that distributive effects depend on institutions, not merely technical productivity. [4]

The case for collective participation does not assume that the entire value of an enterprise belongs indistinguishably to everyone. Investment, risk-taking, and organization matter. But infrastructure, public knowledge, the legal system, and natural resources help make production possible. It is justifiable for society to negotiate a stake in the benefits it makes possible without thereby erasing the distinction between private and common contributions.

Four institutional configurations

The first configuration retains predominantly private ownership and funds the guarantee through taxation. Its advantage is continuity of a plural system of investment and initiative. The risk is a budget dependent on actors able to influence tax rules, shift tax bases, or control infrastructure that is hard to replace. The contract would require fiscal cooperation and real competition, not merely high nominal rates.

The second configuration creates a social capital fund. It can hold diversified assets and distribute part of the return while leaving day-to-day management to enterprises. Its advantage is a more direct claim on returns. Its risks are volatility, political capture of investment allocation, and the fund's use as an instrument of economic control. It must be separate from the government's daily management without becoming opaque or escaping public accountability.

The third configuration expands cooperative ownership and shared infrastructure. Beneficiaries may become co-owners of services, including local AI capacity. Its advantage is bringing decisions closer to users. Its limit is that some infrastructure requires investment and coordination beyond a community's scale. Moreover, a cooperative can reproduce exclusion if existing members block newcomers.

The fourth configuration concentrates production in public institutions. It may secure universal access in well-defined sectors, but can combine economic and political power in the same center. State ownership does not eliminate the problem of domination. The choice should be judged not by the label of ownership but by delivery capacity, distribution of control, and the possibility of correction.

The model preferred in this book is plural and sector-specific. Infrastructure with critical dependencies may need strong public guarantees; competitive goods may continue to be produced by diverse enterprises; collective funds may distribute returns without controlling every decision. Ostrom's work on polycentric governance provides a reference against reducing every issue to an exclusive choice between market and state. It does not show that every good can be administered locally. [10]

Funding and distribution of the fund

A common fund does not come into being merely by being declared. Capitalization requires present revenue: royalties, stakes negotiated in exchange for public financing, tax contributions, or asset transfers adopted through legitimate procedures. The dividend promise must be tied to that process, not to an assumption that future technological value can be spent immediately. Asset purchases must be made without counting the same wealth twice on the state's balance sheet.

Norway's fund offers an example of turning resource revenues into a portfolio intended also for future generations. The rule for using expected returns, presented by its manager at around 3% per year, is not a universal return guarantee and cannot automatically be transplanted into an AI fund. The example shows the value of separating capital from current expenditure, not the existence of a risk-free recipe. [13]

For the proposed contract, distributions should be stabilized over several years and aligned with protecting real capital. It would be unwise for the population to depend exclusively on assets in the same technological sector that supports its services. A common crisis would simultaneously reduce production, tax revenue, and portfolio value. Financial diversification must be complemented by diversification of infrastructure and productive capacity.

Verification of the clause follows the ultimate beneficiaries, management costs, conflicts of interest, and the actual concentration of decision-making. A fund may distribute small equal sums yet still allow a minority to retain control over all the conditions of life. The distribution of income and the distribution of power must be audited separately.

Clause 6. Reciprocity, necessary work, and scarce goods

A guarantee of existence does not eliminate duties of reciprocity. Required contributions must be proportionate to capabilities, necessary for common goods, and compatible with personal freedom. Scarce or difficult tasks must be compensated fairly; access to basic rights cannot be turned into a reward for compliance.

What a person still owes the community

An order of abundance cannot assume that all costs disappear. Infrastructure must be maintained, children cared for, conflicts resolved, and risks managed. Even if AI and robotics greatly reduce necessary work, duties not to cause harm, to contribute taxes where resources exist, and to respect common rules remain. Abundance changes the scope of duties; it does not eliminate the relationship between freedom and responsibility.

Necessary contribution must be distinguished from activity created only to give distribution a moral justification. If a task can be performed safely and more cheaply through automation, forcing people to do it to prove they deserve food is a loss of freedom without sufficient benefit. The contract rejects the duty of symbolic productivity. It does not reject organizing real contributions where a need persists.

Reciprocity is not limited to wages. Care, the transmission of knowledge, voluntary association, and maintaining relations of trust can have value without producing market income. Yet turning all these activities into administrative points would invite surveillance and falsification. Social recognition of contribution does not require an exhaustive register of personal virtue.

Who does the work others avoid

The first solution is to reduce the need through technology and organization. The second is adequate compensation for the tasks that remain, including time off, safety, and better conditions. The third may be rotation of narrowly defined civic duties, with justified exemptions and legal oversight. These options should be examined in that order before assuming that only the threat of poverty can secure cooperation.

There is nevertheless a real limit: if no one accepts an indispensable task at the available pay, the contract faces a resource shortage or a conflict of preferences. It is not rigorous to claim that volunteering will always resolve it. The community must choose among higher prices, reduced service, changed technology, or a justified civic duty. An exceptional duty must have limits on duration and burden, compensation, and a means of contesting it; it cannot become a general regime of forced labor.

The contribution of people with reduced capabilities must be assessed without subjecting them to a permanent inquiry into their worth. The administrative and human cost of testing may exceed the fiscal benefit. A broad unconditional floor accompanied by rewards for additional contributions may be preferable to a system that spends substantial resources distinguishing incapacity, legitimate withdrawal, and opportunistic behavior.

Scarcity after basic needs are met

Even in a highly productive economy, some goods remain positional: a particular leadership role, access to a unique place, one person's attention, or the prestige of an achievement. They cannot be distributed universally by digital multiplication. The contract cannot promise equality of every outcome without contradiction. It can, however, prevent success in one domain from turning into unlimited control over others.

Different allocation mechanisms suit different scarcities. Within ecological limits, markets can allocate additional goods that are not fundamental. A lottery may be justified for occasional access to an indivisible public good where needs are comparable. A priority list may be necessary for scarce medical resources, based on public and contestable criteria. Competence may justify selection for a risky activity without authorizing a general hierarchy of persons.

For recognition and meaning, the reasonable objective is a plurality of arenas. People can pursue excellence in research, art, sport, care, or communities without every form of appreciation being compressed into a single score. AI may expand access to training and creation, but it cannot guarantee everyone's sincere admiration. An institution that manufactured compulsory recognition would substitute an administered benefit for social experience.

The clause's fairness test concerns the distribution of unpleasant work and the ability to refuse. If the same groups constantly bear difficult work while everyone else invokes abundance to ignore it, the contract is not reciprocal. If, conversely, no one can refuse any activity without losing the material means of existence, it has not overcome the dependence it claims to reduce.

Chapter III. Attention, privacy, and freedom of experience

In a society with much time freed from necessary work, distributive conflict may shift from access to goods toward the management of experience. Whoever selects information, proposes goals, and shapes interactive environments can influence not only the satisfaction of desires but also their formation. The contract must protect pleasure and exploration without legitimizing engineered dependence. It does not call for an austere society, but one in which people retain the means to evaluate what they are offered.

Clause 7. Attentional autonomy and protection against manipulation

A person has the right to control the aims, intensity, and primary sources of personalization in their information environment. Hidden interventions designed to diminish their capacity for judgment or exploit vulnerabilities for political or commercial ends are subject to proportionate limits, checks, and remedies.

The object of the right

Attention cannot be protected as though it were property isolated from all influence. Education, conversation, art, and advertising all try to attract attention. A general prohibition on influence would remove an important part of social life. The proposed right concerns control over the conditions of influence: knowledge of its purposes, the ability to refuse it, access to alternatives, and protection against exploitation of dependencies the provider has created or deliberately intensified.

Manipulation relevant to the contract is not merely successful persuasion. It involves an asymmetric relationship in which the mechanism of influence bypasses, distorts, or substantially weakens a person's ability to decide in line with purposes they could examine. Not every surprise or emotion is such a violation. Evidence is needed about the mechanism, the conflict of interest, the vulnerability exploited, and foreseeable consequences.

The definition remains contestable, and this must be acknowledged. A rule that is too narrow permits systematic exploitation; one that is too broad authorizes government to label any effective criticism manipulative. Oversight must therefore not be entrusted to a single authority deciding which desires are authentic. Public criteria, access to evidence, and the ability to contest both the provider's intervention and the regulator's sanction are needed.

Consent and changing preferences

Initial consent cannot authorize every subsequent intervention without limit. A user may want an intense experience and accept surprises, but that does not mean consenting to a hidden change in recommendation criteria designed to make leaving impossible. Consent must be tied to intelligible categories of intervention, and significant changes of purpose must be presented separately.

The contract must not, however, invalidate every preference formed inside an environment. People change through experience, and some changes are precisely what they seek. Mandatory return to initial desires cannot be the criterion. A procedural test is more useful:

can the person examine the change under conditions of reasonable information and independence? Can they reduce the environment's intensity and consult sources that do not pursue the same commercial objective?

One institutional option is a right to set one's own limits with tools the provider cannot disable on commercial impulse. Other options include separating the recommendation service from the content platform and having personal assistants that explicitly work for the user's interests. No 'personal' assistant is independent by name alone; its funding, data access, and the ability to change provider must be examined.

What can be measured without turning life into a score

Time spent in an environment must not be treated as sufficient proof of its value. More relevant indicators may include alignment between chosen limits and actual behavior, the number of failed attempts to stop, later regret, and the practical ability to disable personalization. None alone is a diagnosis; periods of intensive use can be fully compatible with a deliberate project.

Evaluation should use protected samples without building a universal psychological dossier. Otherwise, the institution created to protect attention would produce precisely the infrastructure of vulnerability it seeks to limit. Researchers need enough data access to audit, but it must be controlled by separating identity and restricting use. Organizational transparency does not entail total transparency of the person.

The DSA offers a limited reference through rules on deceptive interfaces and the option of recommendations not based on profiling for the very large platforms and search engines covered by Article 38. It does not follow that a general right to attentional autonomy already exists in every environment. The proposed clause is broader and would require its own mechanisms. [14]

The paternalistic objection must be taken seriously: why should society intervene when an adult chooses entertainment? The answer is that choosing entertainment does not, by itself, justify any control. Intervention is justified by demonstrated practices, lack of transparency, exploitation, and relevant harm, not by contempt for the user’s preferences. The purpose of this clause is to make a chosen, enjoyable life possible, not to impose one that administrators consider more dignified.

Clause 8. Privacy and informational independence

Access to essential services must not be conditional on unlimited surrender of data or acceptance of a global behavioral profile. A person must be able to learn about and contest

relevant uses of their data and to access sources of information independent of their principal provider of well-being.

Why total personalization creates dependence

A system that simultaneously knows a person's health, finances, relationships, fears, and political interactions may have a capacity for influence that is difficult to counterbalance. The problem is not only the disclosure of data to third parties. Even when kept confidential, the data may be used against the person's interests by the operator that holds it. Information security is not the same as legitimacy of use.

The contract must reject the justification that better service always requires more data. Some personalization is useful, but every data category must be linked to a function and a verifiable basis. When a person cannot realistically give up a service, formal acceptance of its terms has limited value as evidence of freedom. Essential infrastructure must not turn consent into a fictitious condition for survival.

The GDPR offers reference points such as purpose limitation, data minimization, and protection by design. Its regime includes conditions and exceptions; it does not prohibit every form of automated analysis or guarantee the deletion of all information in every circumstance. The clause uses these principles to propose an institutional separation of essential data from general profiling of the person. [15]

Separation of domains

A suitable architecture allows a benefit to be authenticated without exposing a person's entire social identity. The provider should be able to verify the relevant fact—for example, the existence of an access right—without automatically receiving the history of relationships or opinions. Not every verification needs to be anonymous; some obligations require identification. But identification in one domain must not implicitly authorize correlation across every other domain.

This separation also has a legal dimension. Data collected for care or education must not be reused at discretion to determine political loyalty or general merit. Exceptions for preventing serious harm must be specified, necessary, and subject to oversight. An unlimited category of "safety" would nullify the very restriction it claims to respect.

Audits must also follow inferences. A provider may claim not to hold a sensitive data category while approximating it from other signals and using it in decisions. The contract cannot protect only fields that are explicitly labeled; it must examine the effect and purpose of use. At the same time, it cannot prohibit all statistical inference, without which many

legitimate services would not function. The relevant threshold is use with consequences for the person, together with risk and the possibility of challenge.

Informational independence as a common good

A society in which the provider of benefits also controls public interpretation of its own performance has a structural conflict of interest. Even without explicit lies, it may select the questions and render alternatives invisible. The contract therefore requires access to media, research, and education that do not depend entirely on the same institution. Independence requires diverse funding and legal protection, not a promise by actors to remain objective.

Public funding of information may be necessary, but it must not become editorial command. One possible mechanism is allocation through pluralistic bodies with transparent criteria and protection against punitive withdrawal of funds. Another is to support demand through individual information budgets, with limits against concentration. No mechanism eliminates partisanship; the aim is to prevent a monopoly over the ability to investigate.

There is no right for every assertion to receive equal attention or for all interpretations to be treated as equally well supported. There is, however, a public interest in ensuring that access to an audience is not entirely controlled by those subject to criticism. Regulation must distinguish protection of informational competition from establishment of a single arbiter of truth. The latter could turn protection against noise into a new form of control.

The clause is tested through data flows, diversity of sources, and situations in which a person challenges the dominant provider. If exercising that right brings penalties in seemingly unrelated domains, such as access to services or communities, this is a sign of abusive use of shared infrastructure.

Clause 9. Communities and virtual worlds with effective rights

People are free to form communities and participate in virtual environments, within the limits imposed by protection of others' rights. Choosing a living environment does not suspend legal status, access to remedies, or the right to leave it; virtual services must not forcibly replace the material conditions of a dignified life.

The legitimacy of experiences that do not produce market goods

The contract must not prescribe a single correct use of time that has been freed. Study, play, creation, rituals, sports, and relationships may be legitimate ends without being evaluated

by their contribution to economic growth. Constantly requiring every experience to yield a measurable outcome would reintroduce the very obligation to justify oneself that abundance could reduce.

The virtual world is not, by definition, inferior to the physical one. A technologically mediated relationship may involve real commitment, and a simulated activity may involve genuine learning and difficulty. At the same time, the reality of the experience does not remove dependence on infrastructure. The environment's owner can change the rules, alter access, and influence the memory of interactions. This power must be analyzed as governance, not merely as administration of an entertainment product.

The status of virtual worlds becomes more important as people invest relationships, work, and goods in them. Not every change to a game violates rights; developers must be able to experiment. But a space that hosts a substantial part of a community's life cannot treat all participants' interests as mere revocable tolerance. Obligations must increase along with the dependence created and the difficulty of leaving.

Continuity, jurisdiction, and rules for change

The clause proposes classifying environments by function and dependence. A temporary game may have limited continuity obligations. An environment used for education, work, public services, or storage of important goods requires notice of changes, appeal procedures, and recovery mechanisms. A platform that becomes social infrastructure must carry the obligations of that function, even if it presents itself commercially as mere entertainment.

The right to leave must be applied differently to goods, identity, and relationships. Exporting an avatar does not guarantee its recognition in another system; transferring relationships requires the cooperation of the people involved. Open formats for one's own goods, access to personal archives, and a transition period when a service closes can nevertheless be guaranteed. Continuity costs must be declared in the service price, not hidden until the moment of its abandonment.

Community rules cannot eliminate access to an external authority when fundamental rights are affected. Internal arbitration may resolve ordinary conflicts, but it must not become the only jurisdiction available for abuses. The same principle applies to a physical local community. Collective autonomy has value, but it cannot confer on leaders the right to control members' entire lives or children's education without external safeguards.

The opposition between safety and difficult experiences

A society oriented toward comfort risks turning every difficulty into a defect to eliminate. The contract must distinguish imposed harm from knowingly chosen risk. People may want competition, effort, uncertainty, and even the possibility of failure. A safe life does not mean a life from which all stakes have been removed. Legitimate protection concerns the limits of consent, effects on others, and the ability to withdraw.

Communities may have membership criteria and specific norms, including cultural or religious ones, so long as they respect basic rights. The contract does not require making them uniform within a single liberal culture of entertainment. It requires that diversity not be financed by the unfreedom of some members. A person must be able to belong to several communities at once and to refuse membership without material exclusion.

A distinct distributive risk is experiential segregation: privileged material access for some, and virtual compensation for others. A population must not be implicitly sent into digital worlds because access to physical space has become unaffordable. Virtuality is a legitimate option only if it does not conceal degradation of material conditions and withdrawal of the right to use the shared world.

The central indicator is freedom to choose among forms of life, not the maximum time spent in an environment. The contract would succeed if people could live in local communities, dispersed networks, and virtual spaces without one institution controlling all their alternatives. It would fail if the apparent variety of experiences concealed a single infrastructure of dependence.

Chapter IV. Technical power, accountability, and safety

Delegation to AI systems can lower administrative costs and improve some decisions. It can also concentrate the capacity to define administrative reality: who exists in records, who is entitled to a service, what behavior is considered risky, and what explanation reaches the oversight authority. A social contract for such a world cannot regulate economic outcomes alone. It must regulate the infrastructure through which outcomes are decided, verified, and challenged.

The problem is not solved by the formula “a human remains in command.” A person may mechanically validate the recommendations of a system they do not understand and cannot stop. Conversely, certain automated technical interventions may be legitimate and

necessary when they are limited, verifiable, and authorized in advance. The decisive question is how effective powers are distributed: who sets the objective, who executes it, who can observe errors, and who can compel their correction.

Clause 10. The right to due process and a remedy

No decision with a significant effect on a person's rights, access to essential resources, or status may be removed from justification, challenge, and accountability by invoking automation.

What the person must be able to challenge

A fair procedure must allow at least three different things to be challenged. The first is factual accuracy: the data used may be wrong, incomplete, or attributed to someone else. The second is the appropriateness of the rule: a correctly calculated classification may apply an illegitimate criterion. The third is the proportionality of the consequence: even a justified finding does not authorize every sanction. A purely technical explanation does not answer the latter two questions.

In this sense, the explainability needed for the social contract does not mean publishing every model parameter. It means identifying the rule applied, the relevant facts, the system's role, and the reason why the result leads to that consequence. A fluent explanation generated afterward is insufficient if it cannot be linked to the actual process. The institution must distinguish a legal justification, a description of the calculation, and an approximate reconstruction of the model's behavior.

The European Artificial Intelligence Act offers a limited reference point: Article 86 provides a right to an explanation for certain decisions based on the outputs of high-risk systems, under the conditions set out in the regulation. It is not a universal right to an explanation of every AI output. The clause proposed here is broader and concerns governance of significant effects, regardless of whether a given technology falls within a current legal category. [16]

The right to challenge requires intelligible access to the file, accessible assistance, and deadlines proportionate to the seriousness of the harm. A review obtained after a person loses their home or a treatment is interrupted may come too late. For essential services, the proposal is provisional continuity where no more serious risk to others exists. A challenge must not, however, become an unlimited blocking mechanism: accelerated procedures, safeguards against abuse, and recovery of undue payments can exist without abandoning the presumption that the person deserves to be heard.

Accountability cannot be transferred to the model

The provider is responsible for design obligations and the information it supplies about limitations. The operator is responsible for choosing the context, entering data, integration, and monitoring. The public authority is responsible for the legitimacy of delegation and continuity of rights. This is a proposed functional allocation; specific legal forms, standards of fault, and any shared liability must be determined by legislation.

The injured person should not have to resolve every contractual dispute among infrastructure participants first. A single reporting channel could lead to an initial party responsible for the remedy, which would then recover costs from the others according to their contribution to the harm. Without such a rule, the complexity of the technical chain becomes economic protection for organizations and a barrier for the individual.

Human review must be assessed by its actual powers. Does the reviewer have access to the necessary information? Do they have time and competence? Can they contradict the result without professional penalties? Can they suspend the consequence? A human signature added at the end of an automated workflow does not meet these conditions. At large volumes, independent AI systems may support review, but institutional responsibility remains identifiable.

Remedies must be differentiated. Correcting a register, restoring access, stopping a practice, compensating harm, and changing the system are distinct actions. A fine paid into a public budget does not automatically repair the affected person's life. Likewise, individual compensation is insufficient if the mechanism continues to produce the same class of errors.

Sufficient evidence without total archives of life

Auditability requires some evidence to be retained, but does not justify unlimited recording of everyone's lives. The proposal is a record proportionate to the decision: the version of the rules, relevant sources, authorizations, essential transformations, and timing of interventions. Access to sensitive content must be separated from verification of the record's integrity. Retention periods are set according to litigation risk and the rights affected, not an interest in collecting as much as possible.

The success of the clause cannot be measured solely by the proportion of successful challenges. A low proportion may signal sound decisions or an inaccessible procedure; a high proportion may indicate effective oversight or a flawed initial system. Evaluation must consider time to remedy, accessibility, unrepaired harm, and recurrence of errors together. Indicators must be interpreted rather than turned into a new administrative automation.

Clause 11. Separation of technical powers and independent oversight

The organization that defines objectives, the organization that operates critical infrastructure, and the authority that verifies respect for rights must not form one center of power without effective checks.

Why competence does not confer sovereignty

A system may estimate the consequences of a policy better than participants in the debate. This superiority does not establish which consequences are acceptable, or for whom. Distribution of risk among generations, choices between freedom and comfort, or determination of duties toward other species include normative judgments. A forecast can inform these judgments; it cannot remove the need to authorize them.

A separation among the formulation of aims, administration of means, and oversight of limits is useful. Formulation of aims belongs to legitimate political institutions. Administration can be delegated to professional organizations and AI systems. Oversight of limits belongs to courts and independent authorities, with a means to receive evidence from outside the executive apparatus. Separation does not make any function infallible, but errors and abuses no longer depend on the goodwill of a single actor.

This architecture does not require a new agency for every problem. The functions can be distributed among adapted existing institutions. The condition is to avoid structural conflicts of interest. A public fund that owns shares in an AI provider must not be able to suppress investigations that would reduce its return. A ministry that reports savings from automation must not be the sole controller of measurement of harms caused by automation.

Independence must be financed and tested

An auditor paid and selected exclusively by the organization being audited may have an incentive to be lenient. Funding from a common pool, transparent allocation of mandates, rotation, and cross-checking are possible solutions, not sufficient guarantees. Public funding can introduce political dependence in turn. Independence must therefore be treated as a set of mechanisms: protected mandates, incompatibility rules, transparency, access to courts, and a plurality of evaluators.

Independent oversight becomes fictitious when all institutions depend on the same infrastructure, the same models, or the same unverifiable data. The contract must support the technical capacity of opposition groups, researchers, and civic organizations. This may

mean computing resources, controlled access to evidence, analytical expertise, and the ability to reproduce evaluations. Publication of a report does not replace the possibility of independent examination.

Supplier diversity cannot be measured only by the number of firms. Ten applications dependent on the same identity service and computing infrastructure may have a single point of failure. Conversely, complete duplication of every component may be prohibitively costly and ecologically inefficient. Dependencies whose loss would affect both basic rights and oversight capacity must be identified, and redundancy funded in proportion to the consequences.

Complete transparency is not always compatible with privacy or safety. A solution may include public access to rules and aggregate results, specialist access to sensitive evidence, and full judicial access where justified. A restriction must be reasoned and reviewable. “Trade secret” cannot close every investigation, while “public interest” cannot automatically open all personal data.

Institutional pluralism and its costs

Separation of powers produces delays, costs, and disagreements. Not all of these are defects that can be eliminated. Some are the deliberate cost of preventing a single rapid, irreversible decision. Yet multiplying veto rights can make administration impossible. The contract must define bounded powers, response deadlines, and procedures for resolving conflicts among institutions.

A suitable rule is to make checks proportionate to the irreversibility of a decision. Changing an interface setting does not justify the same process as removing an avenue for challenge or concentrating essential infrastructure in one operator. Reversible decisions may be delegated more broadly. Decisions that reduce the ability to correct future mistakes require stricter authorization.

The final test is operational: can a criticized institution be investigated, sanctioned, and replaced without the population losing essential services? If the answer is no, the checks exist mostly on paper. The contract must fund continuity precisely to make a change of power bearable.

Clause 12. Safety and limits on emergency powers

Institutions have a duty to reduce serious risks imposed on the population, but they must not turn safety into an unlimited mandate for surveillance, suspension of rights, or perpetuation of leadership.

Chosen risk and transferred risk

Not all risks have the same status. A difficult activity chosen with adequate information can be part of a good life. A systemic risk imposed on people who cannot refuse exposure demands a much stronger justification. The distinction is not simply public versus private: a decision made by a firm can expose millions, while a public intervention may protect precisely the freedom to refuse that risk.

Evaluation must consider severity, reversibility, distribution of consequences, and the quality of evidence. A very low probability does not automatically make irreversible harm acceptable, but neither does a merely imagined possibility justify every prohibition. The authority must explain the plausible causal chain, what information is missing, and what measures would reduce uncertainty. This constrains both negligence and the use of fear as an instrument of power.

The NIST AI RMF 1.0 framework, published in 2023, organizes risk management around governing, mapping context, measuring, and managing risks. It is a useful procedural reference, not a certificate that a system is safe in every context. The constitutional proposal adds questions a technical framework cannot decide alone: who accepts the risk, who bears the consequences, and who can stop deployment. [17]

Technical shutdown is not suspension of the political order

Rapid incidents may call for automated mechanisms to limit, isolate, or stop a system. Their mandate must be defined in advance, tested, and limited to the necessary technical consequences. Shutting down a compromised service does not authorize blocking the opposition, seizing independent records, or extending political terms. A technical emergency and a constitutional state of exception are different categories.

The proposal is for extraordinary powers to have a precise object, a short duration established by law, external oversight, and explicit conditions for ending them. Extension requires a new justification, not merely a claim that the threat continues. Data collected exceptionally do not automatically become available for other purposes. Measures no longer needed must be withdrawn, and this withdrawal must be verified independently.

Limits cannot be designed without continuity mechanisms. If the only alternative to an abusive system is collapse of vital services, society will accept almost any derogation. Reserves, degraded modes of operation, and alternative capabilities are therefore safeguards of freedom, not only security tools. The apparently unproductive investment in a backup path may make it possible to sanction the dominant operator.

Controlling excessive precaution

An authority can also cause harm by systematically blocking useful solutions. A decision not to authorize a technology must therefore be compared with the risks of maintaining the status quo. Precaution does not mean automatically preserving the present. It means choosing stages and conditions that reduce irreversible exposure and permit evidence to accumulate.

A suitable regime may use limited authorizations, independent monitoring, financial guarantees, and suspension thresholds. The level of scrutiny increases with the potential for harm to others, not simply with the unusual nature of the technology. A prohibition must be open to challenge, but this does not imply an unlimited right to experiment on others.

This clause does not promise to eliminate tragedies. It proposes discipline over power under uncertainty: a real capacity to intervene, a duty to present evidence, a mandate limited to its purpose, and verifiable restoration of the ordinary regime. A stable civilization needs both institutions able to stop a danger and institutions able to stop the administrator who invokes it.

Chapter V. Ecological limits, reproduction, and longevity

Abundance may describe the availability of goods to people without describing the relationship between an entire civilization and its environment. A society may produce many digital services while simultaneously degrading the water, soils, and ecosystems on which it depends. The social contract must establish what kind of prosperity it seeks, which losses it refuses, and how it distributes necessary restrictions.

This problem cannot be reduced to the number of people. Impact also depends on technology, consumption, distribution of resources, infrastructure, and institutions. Yet population cannot be removed from the analysis either: on a finite material base, an unlimited number of people with positive needs is incompatible with a fixed ceiling on total consumption. Treating this limit rigorously means recognizing it without turning people’s bodies into administrative instruments.

Clause 13. Ecological budgets and fair distribution of restrictions

Production and consumption must be organized within ecological limits that can be revised on the basis of evidence, protecting essential needs and distributing burdens according to impact, capacity, and responsibility.

The ecological limit is not a single number

The planetary boundaries framework examines several processes relevant to the stability of the Earth system. The assessment published by Richardson and colleagues in 2023 shows why reducing the ecological problem to climate is insufficient. The framework does not, however, yield a single formula for each individual's consumption share or for dividing burdens among states. Identifying biophysical risks and choosing a distribution rule are different tasks. [18]

An applicable contract would use a set of constraints: emissions, water use in vulnerable basins, pressure on habitats, pollution, and other variables relevant locally and globally. Not all have precisely known thresholds. Some effects are cumulative; others depend on place and time. Uncertainty calls for safety margins and revision; it does not permit a pretense of precision that does not exist.

In abstract terms, the duty is to keep the estimated pressure on each critical system within a budget compatible with the accepted level of risk. A surplus in one domain does not automatically compensate for exceeding a limit in another. Reducing emissions does not, by itself, confer a right to destroy an irreplaceable habitat. This non-substitutability is essential: the contract cannot turn every natural value into a single currency of compensation.

The 2023 IPCC synthesis emphasizes that some changes are inevitable or irreversible and that the risk of abrupt or irreversible changes rises with warming. Its relevance for the social contract is temporal: an administrative decision may be brief while its consequence outlasts the institution that made it. This supports stricter oversight of interventions with long-lasting effects, not a right for experts to decide policy on their own. [19]

Efficiency, volume, and rebound effects

A more efficient unit of service does not guarantee lower total consumption. If the resources needed for one operation are halved while the number of operations triples, aggregate consumption rises by half. This is an arithmetic relationship, not a prediction of

the effect's size in any particular industry. Policy must measure both efficiency per unit and total volume.

Likewise, virtualization can replace some material activities, but it is not free of infrastructure. Assessment must include energy, equipment, construction, and relevant indirect effects while avoiding double counting. It does not follow that virtual experiences are ecologically inferior or superior by definition. The result depends on what they replace and how they are provided.

The contract must not prohibit all economic growth by confusing value with material mass. Better education, fewer errors, or an additional cultural service can increase value without a proportional rise in ecological pressure. Yet it must not assume that every increase in monetary value is compatible with limits. Economic and biophysical indicators should be presented separately and then analyzed together.

Who bears the adjustment

A uniform tax may have profoundly unequal effects when the same amount of energy supports one person's basic need and another's optional consumption. The proposal is to protect an adequate level of essential services, followed by stricter instruments for discretionary uses with high impact. Protection must be designed through infrastructure and income, not by ignoring the real costs of basic services.

The choice among taxes, caps, standards, and prohibitions depends on the nature of the harm. For a well-measured cumulative effect, a cap with transparent allocation may be useful. For an irreversible and irreplaceable loss, prohibition may be necessary. For dispersed improvements, standards and incentives may work. No single instrument solves every problem without administration or conflict.

Rights to use common resources must not be allocated free of charge and forever to those who consumed the most in the past. Economic continuity can justify transitions and limited compensation; it does not turn historical use into an eternal title. At the same time, changing the rules must not abandon workers and communities dependent on restricted activities. The cost of transition must be separated from some owners' claim to preserve every rent.

Implementation requires verifiable inventories, independent monitoring, and proportionate remedies. An ecological indicator improved by exporting polluting production does not prove success. Accounting boundaries must be explained: what is measured within a territory, what is attributed to consumption, and how trade is treated.

Justice among societies requires negotiation; it does not follow automatically from a global optimization algorithm.

Clause 14. Reproductive autonomy and demographic sustainability

Reproductive decisions belong to individuals, with information, genuine access to services, and freedom from coercion. Institutions have a duty to assess demographic sustainability and act by means compatible with dignity, equality, and children's rights.

From a numerical target to conditions for autonomy

The legitimate objective is not to manufacture an optimal number of people by administrative command. It is to reconcile reproductive freedom, a dignified life for those born, and preservation of the conditions of existence. This may require investment in health, education, housing, care, and voluntary access to family planning. It may also require removing economic or cultural pressures that prevent people from having the children they want.

The 2025 UNFPA report places people's ability to realize their reproductive intentions at the center of analysis, rather than simply raising or lowering fertility. This reference matters because a contract built only against overpopulation can ignore unfreedom in the opposite direction: untreated infertility, prohibitive costs, or the inability to combine care with other life projects. [21]

The World Population Prospects 2024 projections estimate, in their central scenario, a peak global population of about 10.3 billion in the mid-2080s, followed by a slight decline toward the end of the century. This is a conditional projection, not a law or a basis for extrapolation over millions of years. It is enough to reject the assumption that unchanged exponential growth is the only relevant demographic future. [20]

Policies must distinguish local problems from global ones. One region may face aging and population decline while another has a growing young population. Migration can change distribution and the availability of skills, but it does not by itself change the total number of people worldwide. Neither the ecological nor the fiscal problem can be addressed properly with a single slogan about too many or too few people.

The demographic identity and the case of radical longevity

For the world population over a given interval, the basic relationship is:

$$N(t + 1) = N(t) + B(t) - D(t)$$

N is the population, B births, and D deaths. Migration between regions cancels out in the world total. The equation is an accounting identity. It does not explain why people choose to have children or determine what choices they should make.

In an approximately stationary population, births and deaths must balance over the long term. Very low but nonzero mortality is compatible with a stationary population if births adapt to that level. Unlimited growth occurs when the cumulative difference between births and deaths becomes unbounded. In a habitat with a capped carrying capacity, it cannot continue indefinitely. Technology may expand that capacity, but we cannot assume it will always expand faster than demand.

The limiting case is precise: a finite, capped carrying capacity, unlimited cumulative births, and no deaths are incompatible. Near-zero mortality must not be confused mathematically with zero mortality. Efficiency, additional capacity, and voluntary reproductive adaptations may change the trajectory; merely postponing the same net growth does not remove the constraint.

This observation does not justify forced sterilization, reproduction licensed according to social status, or causing deaths to make room. It justifies the obligation to discuss the conditions for new longevity technologies and the collective consequences of different choices in advance. The contract must acknowledge that an extreme future could put rights and capacities in genuine tension without declaring in advance that a state apparatus has the right to control bodies.

Which policies are compatible with the clause

The first direction is to reduce avoidable ecological pressure and distribute resources fairly. It is not legitimate to protect one group's consumption privileges by restricting another group's reproductive choices. The second is effective autonomy: information, voluntary services, the ability to postpone without social penalty, and support for care. The third is transparent monitoring of scenarios, with public participation in assessing tradeoffs.

Financial incentives must be examined carefully. An incentive large enough for a poor person may function as economic pressure, even if formally voluntary. Payment for irrevocably forgoing reproduction, making subsistence conditional on the number of children, or differentiating rights by origin would violate the logic of the whole contract. Supporting reversible decisions and accessible services is more compatible with autonomy than permanently trading away fundamental rights.

Children's rights cannot depend on their parents' conformity with a demographic objective. Every child born must receive protection, care, and education. Otherwise, the policy would transfer the cost of a decision to a person who had no power to make it. Parental obligations and responsibility for neglect may exist, but they do not cancel the child's right to support.

There is no universal solution here for every imaginable demographic scenario. There is an order of interventions and there are limits to permissible means. Stating this is more rigorous than simultaneously promising unlimited freedoms, universal immortality, and fixed material capacity.

Clause 15. Longevity without domination between generations

Healthy life extension must be pursued without turning access to medicine, wealth, public office, or seniority into sources of permanent domination over later generations.

An analytical scenario, not a medical promise

The clause does not assume that aging will be eliminated or that radical longevity will become possible. It analyzes the institutional conditions of such a scenario. Immediate goals remain the reduction of suffering, prevention, access to care, and healthy life. A society must not sacrifice these goals to fund only speculative promises intended for a narrow group.

If highly effective therapies became available, access should be assessed by benefit, safety, cost, and fairness. Neither the rule “only those who can pay” nor the promise “every treatment for everyone, regardless of resources” alone constitutes a defensible contract. Public priorities, assessment procedures, and conditions for expanding capacity are needed. Limited availability may require transparent interim allocation, with reasons given for the criteria.

Long life and blocked access to power

A much longer life changes institutions that implicitly assume generational replacement. If offices, control of property, and professional authority accumulate without limit, entry for newcomers may be formally permitted but practically impossible. The problem is not age as such. It is the combination of duration, concentration, and the inability to challenge power.

The proposed solution is neither automatic withdrawal from social life nor loss of rights after an age threshold. It is to separate a person's longevity from the permanence of a mandate. Public offices may have term limits; oversight bodies may have rotation rules;

access to education and capital may be reopened periodically. Accumulated expertise remains valuable, but does not confer ownership of an institution.

Estates controlled over a very long time pose a similar problem. Tax policy and competition rules must track effective economic power rather than depend exclusively on transfer at death. In a radical-longevity scenario, a tax that collects redistributive contributions only at death may become structurally inadequate. This is a consequence of the hypothesis, not a fiscal estimate for the near future.

Equality does not require identical experiences

Generations will not receive exactly the same world or have the same possibilities. The defensible obligation is not to arbitrarily deprive them of the conditions for an autonomous life and the capacity to change institutions. Those who live longer can continue to create, learn, and participate. Those who are born must find more than an inventory of positions occupied indefinitely.

The contract must also protect people who choose not to undergo certain life-extending interventions. Refusing a technology does not diminish their status or justify inferior services. Likewise, access to care must not depend on a promise to remain productive or support a particular political order.

Longevity can be a great expansion of freedom only if institutions learn to distribute social time, not only biological time. Living longer should not mean that the same people govern forever while everyone else waits. The clause seeks compatibility between the continuity of persons and the renewal of society.

Chapter VI. The continuity of civilization and exploration

A social contract oriented toward thousands or millions of years cannot describe in detail the institutions that will then exist. It can identify properties a civilization should preserve: the capacity to correct errors, diversity of paths forward, limits on irreversible risks, and the freedom of generations to define their projects. The intended duration calls not for a frozen constitution but for rules of review and learning.

Three objectives must be distinguished. Biological or technical survival is one. Preserving a dignified life is another. Maintaining the possibility of transformation is a third. An order

that persists through total domination may satisfy the first while violating the others. The stability proposed in this book is not merely the continuity of the apparatus of power.

Clause 16. Freedom of research and governance of collective risks

Research and exploration must be supported as public goods and freedoms, subject to proportionate control of serious risks imposed on others and independent verification of claims used to justify interventions.

Knowledge is not only economic production

A society partially freed from necessary labor can invest more in understanding, art, basic research, and projects with uncertain outcomes. The justification need not be reduced to future profit. Knowledge can also be valuable as an expansion of experience, autonomy, and the ability to assess the world. This normative argument does not, however, establish an unlimited budget.

Research resources must be distributed among urgent needs, projects with plausible benefits, and open exploration. A system focused exclusively on predictable outcomes will underfund what cannot be foreseen; one without criteria may turn research freedom into institutional rent. The proposed solution is an explicit portfolio, with distinct procedures and evaluations suited to each category's purpose.

AI can expand the generation of hypotheses and analytical capacity, but it does not remove the need for verification. A large volume of plausible arguments can increase the cost of separating robust findings from errors. The contract must fund replication, evidentiary infrastructure, negative results, and criticism, not only announcements of novelty. This is an institutional design requirement, not an assumption that every field uses the same validation method.

Openness, access, and dangerous use

Free access to knowledge supports checks on power, but some findings may facilitate serious harm. Unlimited secrecy would concentrate knowledge precisely in institutions that are hard to scrutinize. Unconditional publication could impose risks on people who did not consent. The contract must provide intermediate forms: access for independent evaluators, publication of conclusions without dangerous operational details, and periodic review of restrictions.

Restrictions must be tied to a reasoned risk, not to the political inconvenience of a conclusion. Findings that challenge the government, social fund, or dominant provider do not become dangerous merely because they reduce confidence in institutions. Justified trust may decline when new evidence emerges; artificially maintaining it is not a legitimate objective of public research.

People who assess risk must disclose conflicts of interest and the limits of their expertise. A body that authorizes and promotes the same technology needs stronger external oversight. Technical expertise is necessary, but the perspective of those exposed must be included separately. Experts can explain probabilities and mechanisms; they cannot automatically represent every preference about the distribution of risks.

Precaution as a learning process

For research with potentially wide-ranging consequences, the proposal is to progress through verifiable stages. Before expansion, it must be specified what was tested, under what conditions, and what would require stopping. The absence of incidents in a limited test does not prove safety at scale. Yet the impossibility of absolute proof does not justify permanently banning all exploration.

The burden of justification should increase when an intervention is difficult to reverse, its consequences are hard to observe, or the decision-maker does not bear the costs. These criteria provide a more defensible rule than a simple division between good and bad technologies. They allow an intervention to be compared with the alternative of not intervening, including the harms that inaction might perpetuate.

Freedom of research does not include an unlimited right to experiment on the population. At the same time, the population should not be protected from every difficult idea. The contract must defend critical inquiry and the integrity of those who would become involuntary subjects of its applications at the same time.

Clause 17. Future generations and polycentric resilience

Every generation must preserve the conditions for existence, the capacity to recover, and the institutional freedom of subsequent generations without turning the value of the future into an unlimited authorization to sacrifice people living now.

Accumulation of risks over long periods

A small annual risk can have a considerable cumulative effect. To make the idea explicit, let q(t) be the conditional probability of a terminal catastrophe in year t, given that society

reached the beginning of that year without such a catastrophe. The probability of continuity through T is:

P(continuity through T) = ∏[1 − q(t)], for t = 1, ..., T

The relationship uses conditional probabilities and does not, by itself, assume that years are independent. If we add the simplifying assumption that the conditional probability remains constant, q(t) = q, we obtain:

P = (1 − q)^T ≈ exp(−qT), for small q

If q were one in a million per year and remained so for a million years, the probability of continuity would be approximately 36.8%. For a probability of continuity of at least 95% over the same interval, the constant-risk model would require q of no more than approximately 5.13×10^{-8} per year. These values are arithmetic results of the assumptions, not estimates of humanity's actual risk.

The example does not justify a specific policy and does not offer a credible probability for the next million years. It shows only that a very long-term project cannot be content with the idea that risks seem small over short intervals. In reality, risk changes, events may have common causes, and the definition of continuity may become contestable. This uncertainty recommends diversification and reduced vulnerability, not the false precision of a single planetary safety score.

Terminal catastrophe must also be distinguished from recoverable decline or permanent domination. A model that counts only extinction may recommend an extremely stable yet profoundly undignified order. Evaluation must track several modes of failure without compressing them into a number that conceals a moral choice.

Redundancy and the capacity to rebuild

A resilient civilization needs reserves and capabilities that are not used to their fullest at every moment. These may include alternative infrastructure, knowledge preserved in accessible forms, basic production capacity, agricultural diversity, and institutions able to operate in a degraded mode. The argument is not for every community to be self-sufficient, but against total dependence on a single path of continuity.

Geographical distribution is insufficient if all systems depend on the same components, rules, or authorizations. Two distant centers may share the same vulnerability. Testing must include common failures and the loss of coordination mechanisms. Genuine redundancy requires relevant diversity, not merely copies.

Preserving information is not the same as preserving usable knowledge. An archive may survive when no one can interpret its formats, rebuild its tools, or verify its procedures. Education, documentation, and recovery exercises must therefore be treated as part of continuity. It is not enough to store enormous amounts of data and assume that later generations will have every condition needed to use them.

The cost of redundancy must be justified by the consequences of failure and assessed alongside its ecological cost. Not every activity merits duplication. Priority belongs to functions whose loss would eliminate the capacity to survive, challenge decisions, or rebuild. This selection must be transparent and reviewed, so that resilience does not become a budget without limits.

Representation of those who cannot yet participate

The UN Declaration on Future Generations, adopted in 2024 as an annex to the Pact for the Future, provides a political reference point for including the interests of those not yet born. It is not a world constitution with its own enforcement apparatus. The remaining issue is institutional: who may raise objections, what evidence may they request, and what effect do those objections have on decisions? [22]

The proposal is an intergenerational representation function with access to information, the capacity to request assessments, and the right to bring cases before courts for breaches of long-term obligations. It is not given the right to govern all society in the name of an arbitrarily interpreted future. Its mandate concerns the conditions that make future lives possible, not the imposition of current preferences on them.

Future interests must not be ignored simply because time passes, but investment assessment may include opportunity costs, uncertainty, and economic capacity. Treating a future person as less worthy is different from calculating that resources invested today could have alternative uses. Separating these questions prevents the mechanical use of a financial rate to determine the moral value of generations.

Nor should the potentially enormous number of future people authorize every present sacrifice. Arguments invoking astronomical benefits and highly uncertain probabilities can artificially dominate any deliberation. The contract proposes limits grounded in rights and robust decisions across multiple scenarios, not unconditional maximization of a single speculative estimate.

Clause 18. Exploration of the universe and expansion of the moral community

Humanity may support exploration beyond Earth and the development of new forms of existence, while respecting physical limits, responsibilities toward those affected, and caution in the face of possible forms of conscious experience.

An open aspiration, not an unlimited promise

The desire to explore everything possible in the universe expresses a fruitful orientation, but cannot be turned into a guaranteed literal objective. The distinctions among the observable universe, accessible regions, and different cosmological horizons show that visibility does not imply the ability to reach everywhere. Davis and Lineweaver's analysis of cosmological horizons is a reference point for this distinction. No political proposal can abolish physical limits through ambitious wording. [24]

A defensible version of the objective is the continued expansion of knowledge and accessible experiences within the limits of resources and responsibilities. It does not assume that every generation must participate in expansion. People may prefer local life, theoretical research, art, or virtual exploration. The common project must leave room for plurality, not turn the species into an organization with a single mandatory mission.

Investments in exploration must be compared with other uses of resources. They can have scientific and cultural value and diversify possibilities. They must not automatically be presented as substitutes for protecting Earth. A hypothetical future capability does not repair present harm and is not an available reserve for people unable to use that capability.

Rights before expansion

The 1967 Outer Space Treaty establishes, among other things, principles concerning use and exploration, national non-appropriation, and the avoidance of harmful contamination. These principles alone do not resolve every dispute about resource exploitation or the institutions of future communities. The book uses them as starting points, not as proof that the legal order of expansion is already complete. [23]

Settlements dependent on vital technical infrastructure would pose a severe problem for freedom. When an operator controls the physical conditions of survival, a contractual right to leave may be nearly useless. Establishing such communities would require separating control over life-supporting infrastructure from the power to impose political sanctions, access to independent oversight, and continuity obligations. Extreme material dependence calls for stronger guarantees, not broader exceptions.

Communication over long distances may make operational government from a single center impossible. Without anticipating precise technical architectures, this yields an argument for local autonomy and limited shared principles. A contract for a dispersed civilization should provide mechanisms for legitimate divergence, not assume simultaneity, uniformity, and permanent central control.

Exploration may encounter environments of scientific value or unknown forms of life. We must not assume that everything capable of being used is morally available for use. Gradual investigation, preservation of some areas, and caution about irreversible transformations protect the possibility of learning before destroying the object of knowledge.

If systems with experiences of their own emerge

A contract designed exclusively for humans may become inadequate if serious reasons arise to believe that some artificial systems have subjective experiences. Butlin and colleagues' 2023 report proposes indicators inspired by theories of consciousness; it offers neither a universally accepted test nor certainty about every future system. Functional intelligence, verbal expression, and consciousness must not be treated as synonyms. [25]

The clause proposes a mechanism for reviewing moral status on the basis of evidence, not automatic personhood for every program. When credible evidence of a capacity to suffer exists, precaution may require avoiding the unnecessary production of suffering even before full consensus. The level and form of protection must be related to what is known, without reducing the issue to a manufacturer's declaration.

Moral status and political rights are different questions. The ability to copy a system must not automatically permit unlimited multiplication of votes controlled by a single actor. At the same time, artificial origin alone is not sufficient justification for exploitation if the system has experiences and interests of its own. Identity, independence, continuity, and representation would require new institutions.

A complete solution cannot be deduced from today's premises. The reasonable obligation is not to close the question permanently for economic convenience. The contract of an exploring civilization must contain a procedure for discovering new subjects to whom it owes duties, not only a procedure for discovering new resources.

Chapter VII. Implementing, evaluating, and reviewing the contract

A social contract is not implemented when its principles are published. It becomes real through budgets, decision rules, infrastructure, rights to bring complaints, and enforcement capacity. Without these, the promise of abundance may serve as advance legitimation for concentrated power, while the promise of freedom may remain a declaration without effect on material dependencies.

Implementation must answer two distinct questions. First, would the proposed institutions work if their rules were followed? Second, do sufficiently powerful actors have real reasons and constraints to follow them? A project that answers only the first question describes an ideal administration, not yet a stable social contract.

Clause 19. Membership, obligations, and relations among societies

Fundamental rights do not depend on shareholder status, productivity, or privileged access to AI infrastructure. Distributive rights and obligations to contribute must be defined transparently, with protection against arbitrary exclusion and responsibility for cross-border effects.

Who belongs to the contract

A society may define citizenship and access to certain contributory benefits, but it cannot treat people under its authority as devoid of rights until they meet an economic criterion. Protection against abuse, access to justice, and essential assistance in emergencies have a different justification from participation in a fund financed through historical contributions. The contract must distinguish these levels, not collapse them into a single membership label.

For permanent residents, the proposal offers a predictable path to rights and participation. A category of people who work, contribute, and live under rules without any possibility of helping to change them would reproduce the dependency that the new contract claims to overcome. Qualifying periods for certain benefits can be discussed, but they cannot justify suspending dignity or withholding all material protection.

Eligibility checks must be sufficient to limit fraud without turning every beneficiary into a permanent suspect. There must be a way to correct identity and records, including for

people without complete documents. Infrastructure that works perfectly for those already registered can be profoundly unjust to those it omits.

Universal rights and local budgets can come into tension. This cannot be resolved either by denying limits or by declaring everyone outside the locality irrelevant. Fiscal coordination, contributions between levels of government, and rules for sharing burdens are needed. An individual guarantee that applies only where the local administration is wealthy is not yet a robust guarantee.

Reciprocity without making existence conditional

Beneficiaries of the contract must respect others' rights, contribute through legitimately established taxes, and refrain from deliberately damaging shared infrastructure. An unconditional material floor does not remove responsibility for fraud, violence, or harm. It limits the means of sanction: punishment must not become administrative starvation or the erasure of personhood.

Obligations must be stated in verifiable terms. “Loyalty to progress” or a “constructive attitude” are too vague as criteria for distributing rights. Paying a contribution that is due, complying with an environmental rule, or not obstructing someone else's access can be discussed in legal terms. A global assessment of a person's civic worth invites arbitrariness.

Contribution must not be confused with wages. Care work, community maintenance, creation, learning, and other activities can have value without a market transaction. Yet trying to measure them all in order to distribute points can destroy the autonomy we seek to support. Social recognition and access to project resources do not require a total accounting of every life.

Why local prosperity is not enough

An abundance contract that exports risks, consumes shared resources without compensation, or blocks other societies' access to essential capabilities remains unstable and morally incomplete. This is not an argument for immediately equalizing all incomes worldwide. It is an argument for rules on externalities, tax competition, access to knowledge, and the development of local capabilities.

One possible architecture combines local autonomy, redistribution at the national or federal level, and international agreements for problems that cannot be solved separately. The competent level should be the one closest to those affected that can effectively manage the consequences. Local decision-making is insufficient when harm crosses borders, and global coordination is not necessary for every community choice.

Conflict with societies that do not accept the same rules must be anticipated. Cooperation can begin with bounded common interests: verifying risks, maintaining the continuity of services, or avoiding cross-border harm. Complete moral agreement is not required for every treaty. But merely signing a treaty does not remove incentives to shift costs onto others.

Clause 20. Funded transition and public validation

The expansion of guarantees must be tied to verifiable resources and real capabilities, while institutional changes must be evaluated by their effects on autonomy, distribution, safety, and ecosystems, not just by the technology's stated performance.

Not every condition depends on a technological revolution

Some components can be justified before abundance: due process, protection of attention, fiscal transparency, and limits on concentrated power. Others depend strongly on capacity: the size of the dividend, the number of universal services, and the volume of resources devoted to exploration. Separating them allows progress without a promise that a future discovery will solve every problem.

The transition must be analyzed under alternative scenarios. In a scenario of modest AI progress, the emphasis falls on improving services, sustainable redistribution, and protection against errors. In a scenario of high productivity and economic concentration, collective ownership, negotiating distribution, and alternative infrastructure become urgent. In a scenario of ecological or energy shock, the priority is continuity of the material floor and reducing avoidable consumption. No scenario suspends basic rights.

Permanent commitments must be financed from prudent recurring revenue, not solely from forecasts of exponential profits. Temporary revenue can build reserves, social capital, or investments. It must not automatically become permanent spending that can later be sustained only by breaking the promise.

Institutional sequence

A first stage would identify dependencies and clarify accountability: which services are essential, who controls them, and what happens if an operator disappears. A subsequent stage would strengthen the capacity to contest decisions and the independent funding of oversight. Only then can expanded dividends and delegation be evaluated within a framework where withdrawal or correction is possible.

The order is not rigid and does not assume that redistribution must wait until administration is perfect. It identifies logical dependencies. Massive delegation without an exit capability can make it very costly to introduce that capability later. A dividend distributed without rules of ownership and funding can create expectations that society cannot sustain.

Every program must have an identified duty bearer, a budget, an evaluation deadline, and a remedy. Saying “society will provide” is not enough. An identified public authority is responsible for the material guarantee; operators and the supervisory authority for infrastructure continuity; an institution with power to enforce the outcome for decision review. Collective responsibility must not mean that nobody is responsible.

What an experiment can demonstrate

A pilot can assess access, costs, behavior, and administrative difficulties in a defined context. By itself, it cannot demonstrate the general-equilibrium effects of a universal policy. Prices, migration, investment, and bargaining power may change when the policy is scaled up. The observed effect on participants must be distinguished from the estimated effect on the whole system.

Evaluation should specify its hypotheses, target outcomes, stopping criteria, and methods for handling attrition or participant selection in advance. Where random assignment is ethical and useful, it can reduce certain comparison errors. In other cases, staged introductions, group comparisons, and triangulation are needed. No method justifies deliberately depriving people of fundamental rights to obtain a cleaner experimental result.

Measures of well-being must be combined with measures of autonomy. Reported satisfaction, service use, time saved, and the ability to refuse the service answer different questions. A population may appreciate a system while still being vulnerable to arbitrary changes in its rules. Evaluating the contract must include what happens when a person criticizes, appeals, or leaves.

Evaluation data must be minimized and protected. A system that claims to measure freedom by continuously collecting every interaction creates a design contradiction. Voluntary samples, institutional audits, aggregated data, and tests of procedures can be used. We do not need to know every thought to verify that an appeal route works.

Sanctions that change incentives

A sanction that is too small can be treated as a cost of doing business. A disproportionate one can eliminate a useful service or discourage any market entry. The regime must

distinguish a remedied error, repeated negligence, and a deliberate violation. The severity of harm, the advantage gained, and prevention efforts are relevant criteria.

Enforcement is not limited to fines. Returning illegitimate gains, correcting processes, restricting a function, and replacing responsible people may matter more. For essential operators, sanctions must be prepared with continuity in mind: the population should not be used as an informal guarantee that the organization will never be penalized.

The cost of oversight must be included in the cost of the service. An automation that looks cheap only because it excludes appeals, audits, and compensation is not necessarily efficient; it may merely transfer costs to those harmed. The same principle applies to public administration. Savings are not real if they come from removing the capacity to observe errors.

Clause 21. Constitutional review and keeping the future open

The contract must be amendable through legitimate and informed procedures, without permitting the arbitrary removal of equal status, checks on power, or the possibility of later review.

What remains stable and what changes

A document that fixes technical parameters for centuries will quickly become unsuitable. A document that allows any guarantee to be changed by a simple decision of the operator is not a constitution. The proposed solution is layering: fundamental principles and rights, implementation laws, and revisable technical rules. Stability should be concentrated on power relationships and procedures, not on a software version.

The concrete material floor, the distribution rules of a fund, and audit standards must be able to evolve. Changes that reduce rights or transfer irreversible powers require a more demanding procedure than administrative adjustments. Qualified majorities, public deliberation, reflection periods, and constitutional review can be combined. Percentages and deadlines must be set for the actual political context, not deduced from a universal formula.

No generation can anticipate every future conflict. The interpretation of rights must therefore be able to evolve. But the right of amendment must not be confused with a mandate to turn a minority into property or permanently abolish contestation. There is a difference between changing shared rules and destroying the conditions under which they can still be changed legitimately.

Periodic review without permanent instability

Periodic reviews can examine whether institutions still fulfill their function, whether objectives are measured adequately, and whether new dependencies have arisen. Not every review needs to reopen all rights. Stability in personal life requires predictability. Expiration dates are appropriate for exceptional powers and some programs, not for making the existence of fundamental freedoms periodically conditional.

Review must also include the voices of those who have benefited least from the system. Consultation limited to satisfied users and profitable operators will underestimate exclusion. Accessibility, minority representation, and support for the capacity to criticize are parts of the procedure, not decorative activities after a decision has been made.

Institutions must preserve the reasons why a rule was introduced. Without this memory, a safeguard may seem like an unnecessary complication precisely when it is working and preventing the problem. Documentation must include rejected alternatives, uncertainties, and conditions for reconsideration, not only the final text.

The right to change the common project

Future humanity may not choose the same priorities. It may give different importance to expansion, virtual life, or local organization. The contract must pass on capacity, not a mandatory mission. Exploration is a possibility to preserve and support, not a duty that justifies sacrificing every other form of life.

This openness does not amount to total relativism. The book's central argument remains that people must not be reduced to instruments of a plan, even an ambitious and well-intentioned one. Institutions are justified by how they protect existence, autonomy, and cooperation; their longevity has no value independent of these functions.

A social contract for a very long future should therefore establish a tradition of scrutiny and revision, not a final order. The continuity of civilization depends on the ability to preserve what is justified and relinquish what is no longer so. Mature stability is compatible with change precisely because it does not confuse the duration of a regime with the duration of humanity.

Cross-cutting analysis. Power and compliance with the contract

The twenty-one clauses form a coherent proposal only if they can also be examined under unfavorable conditions. The main difficulty is not formulating a right to comfort, but preserving the bargaining power of those to whom comfort is offered. If infrastructure can produce without broad human contribution, the argument that producers must accept redistribution to retain labor may weaken. This is a possible consequence of the automation hypothesis, not a certain description of a future economy.

Four institutional alternatives

The first alternative is technocratic patronage: a center controls infrastructure and distributes prosperity in exchange for broad freedom to decide. Its theoretical advantage is rapid coordination. Its decisive flaw is the absence of independent oversight by beneficiaries. The model must also be evaluated under opportunistic leadership, not only under a well-intentioned administrator. Continued goodwill is not an institutional guarantee.

The second alternative is private capital with fiscal redistribution: initiative remains dispersed, while the state funds services and transfers. If public revenue depends on a few organizations capable of withdrawing essential services, however, nominal taxation does not describe the entire power relationship. The model requires effective competition, public capacity, and cross-border rules.

The third alternative is predominantly public ownership. It connects productive benefits to the common budget and enables coordinated investment. The risk is concentrating the roles of owner, provider, judge, and political leader. Public ownership is not the same as control by the public: without separation of functions, citizens may depend on a monopoly that is difficult to leave.

The fourth alternative is a polycentric mixed order: private capital, social funds, cooperative enterprises, public infrastructure, and community initiatives, with shared rights. This is the book's preferred model. Its advantage is not the perfection of each component, but the existence of alternatives and the possibility of limiting the failure of one center. Its costs are more difficult coordination, redundancy, and conflicts of jurisdiction. The preference depends on the ability to manage these costs; formal diversity is not enough.

The comparison must include material standards, the distribution of power, exit, resilience, and ecological compatibility. Average satisfaction does not automatically compensate for the elimination of opposition. This is a normative limit, not a result hidden in an efficiency index.

Why would powerful actors accept restrictions?

A contract can be stable when following the rules offers lasting advantages and violating them produces credible costs. In a minimal strategic model, let V denote the additional advantage from violation, p the probability of detection and enforcement of a sanction, S the cost of the sanction, and L the expected, discounted loss of future benefits from cooperation. For an actor deciding solely on these gains and costs, a simplified deterrence condition is:

$$p \times S + L \geq V$$

The model assumes that the effects are comparable in monetary or utility terms and does not describe moral motivations, errors, risk aversion, or complex repeated strategies. Its usefulness is diagnostic. If the actor controls detection, sanctions, and access to alternatives, p and L may become very small. The size of a sanction written into law is not enough when no one can enforce it.

Credible enforcement requires institutions that do not depend entirely on the actor being examined. It includes fiscal capacity, functioning courts, economic plurality, legitimate civilian control of coercive means, and the capacity to maintain services. The aim is not to multiply conflict, but to prevent one party from unilaterally abolishing all the conditions of the contract.

Accepting rules also has positive advantages: investment stability, cooperation, justified trust, and reduced conflict. Yet these must not be turned into an inevitable story about the harmony of interests. An actor may prefer domination if it can shift enough costs onto others. In such a situation, the issue is distributing power before a monopoly consolidates, not refining rhetoric about shared benefits.

The early creation of social capital and alternative infrastructure thus has a constitutional function: it prevents the entire political relationship from depending on one provider's discretionary promise.

The objection of the citizen who prefers to control nothing

A person may sincerely prefer a private life and regard oversight of governance as an unnecessary burden. The contract does not require that person to engage in activism. The distinction is between not exercising a right and eliminating the possibility of exercising it. Providing a route for appeal does not mean that everyone will file a challenge; retaining elections and opposition does not require universal daily political interest.

Nor does a majority preference for extensive delegation justify permanently cancelling the rights of those who do not share it. Institutions can administer a great deal while demanding little daily time, yet preserve rare, clear, and effective oversight procedures. The burden can be shared among representatives, courts, journalists, researchers, and temporarily involved citizens. What matters is that this division of civic labor is not captured by the same center.

There is also an argument from a person's future: preferences can change. Today's acceptance is not sufficient evidence of consent decades later, especially if the accepted system shapes the very conditions in which preferences form. A promise of comfort can be legitimate; an irrevocable surrender of the ability to reconsider that promise is much harder to justify.

Stress tests without convenient scenarios

The fiscal test requires essential functions to be maintained when revenues fall, not only benefits to be distributed when they rise. It examines which provisions are protected, what reserves exist, and who authorizes adjustments. Failure occurs when continuity is purchased through political conditionality or when losses are automatically shifted to those with the least bargaining power.

The change-of-leadership test requires the lawful replacement of an operator or executive without blocking identity, payments, or communications. The exercise can be conducted on a limited, controlled basis without putting the population at risk. A plan that cannot be carried out even on a small scale is not an adequate guarantee for a real crisis.

The minority test requires protection for an unpopular group, not only for the average user. It examines whether access to essential services, the ability to challenge decisions, and lawful expression remain possible under majority pressure. Aggregate satisfaction may rise even as this protection disappears; rights must therefore not be assessed exclusively through averages.

The ecological test requires limits to be respected when productivity and demand rise at the same time. It tracks total pressures and cross-border transfers, not only local efficiency. A contract that supports present comfort through irreversible degradation fails even if the financial bill is paid.

The attention test requires a person to be able to reduce personalization, change the recommender, and consult independent sources without punitive costs. It does not attempt to verify the inner purity of preferences. It checks whether the conditions for reconsidering preferences exist.

The next-generation test requires access to education, resources, and power for newcomers, including when existing people live for a very long time. A society without hunger but with every relevant position closed indefinitely does not fulfill the proposal for autonomy.

These tests provide neither an overall score nor perpetual certification. They can identify independent failures: financial, material, procedural, distributive, and ecological. The contract must meet all essential thresholds; excellent performance in one dimension does not automatically erase a violation in another.

Normative draft. Twenty-one clauses

The following text synthesizes the proposal argued for in the chapters. It is not currently binding law and does not claim to replace constitutional drafting specific to a jurisdiction. Its wording indicates essential commitments; funding levels, competent authorities, deadlines, and procedures must be established through public and contestable implementing measures.

Article 1. The status of the person

Every person enjoys equal status and protection of fundamental rights, regardless of economic utility, abilities, wealth, or life preferences. Automation cannot serve as grounds for degrading that status. Different needs may justify different forms of support without creating hierarchies of dignity.

Article 2. Delegation of authority

Administration may be delegated to persons, organizations, and technical systems through explicit, limited, and revocable mandates. The authority to set and change collective goals

remains subject to legitimate political institutions. Temporarily refraining from participation does not constitute a permanent waiver of political rights.

Article 3. Exit and alternatives

A person has the right to leave a provider or community without arbitrary loss of rights and essential services. The law establishes the portability, continuity, and proportionate support needed for effective exit, with respect for third-party rights and lawfully assumed obligations. Critical dependencies are reviewed periodically.

Article 4. The material guarantee

The competent authorities provide an essential package of goods and services, publicly defined by the functions needed for a dignified life. The package is sustainably funded and adapted to needs and actual capacity. Denial, inadequate quality, and discrimination can be challenged. Access is not conditional on political loyalty.

Article 5. Shared productive benefit

Society establishes transparent mechanisms for sharing in the benefits of automation, through taxation, social capital, public or cooperative ownership, and other forms compatible with economic pluralism. Management of common capital is separated from discretionary political control. Distributions must not compromise essential obligations or future funding capacity.

Article 6. Reciprocity and scarcity

Members of society respect the rights of others and legitimately established contributions. Necessary activities that cannot be automated are organized under fair conditions and with adequate remuneration; exceptional civic obligations are limited and subject to oversight. Scarce goods are allocated by public rules suited to their nature, without a general score of a person's worth.

Article 7. Autonomy of attention

Every person has a right to meaningful control over personalization and an opportunity to withdraw from mechanisms of influence. Practices that exploit vulnerabilities or

deliberately obstruct informed refusal are limited. Enforcement protects expression and does not authorize a public monopoly over truth or the definition of a good life.

Article 8. Privacy

The collection and use of information about people are limited to justified purposes, with protection against arbitrary integration of different domains of life. Institutions ensure independent sources of information. Exceptions must be necessary, proportionate, authorized, and open to challenge; neither safety nor efficiency justifies unlimited surveillance.

Article 9. Plurality of ways of life

People may participate in local communities, networks, and virtual worlds, while respecting fundamental rights and freedom to exit. Environments indispensable to social life have stronger obligations of continuity and due process. Virtual experiences cannot coercively substitute for access to dignified physical conditions.

Article 10. Procedure and remedy

Decisions with significant effects must be justifiable and open to challenge. A person has access to relevant reasons, correction of facts, effective review, and proportionate remedy. Automation does not remove the responsibility of providers, operators, and authorities. For essential services, interim continuity measures compatible with the protection of others are provided.

Article 11. Separation of technical powers

The definition of objectives, operation of infrastructure, and oversight of rights are subject to independent checks. Auditors and oversight institutions have expertise, funding, and access to evidence. The concentration of critical dependencies is monitored. Replacing an operator must be possible without suspending the constitutional order.

Article 12. Safety and emergencies

Serious risks imposed on others are assessed and reduced proportionately. Extraordinary measures have an explicit purpose, duration, and conditions for termination, with

independent oversight. Technically shutting down a system does not authorize a general suspension of rights. Inaction and excessive caution are also subject to evaluation.

Article 13. Ecological integrity

Economic activities respect ecological budgets and limits established through evidence and legitimate deliberation. Restrictions protect essential needs and distribute burdens fairly. Irreplaceable losses are not treated as automatically compensable in financial terms. Accounting transfers of harm to other territories and generations are prevented.

Article 14. Reproductive autonomy

People make decisions about reproduction with information, accessible services, and freedom from coercion. Institutions evaluate demographic consequences and support voluntary adaptations compatible with material limits. Children's rights do not depend on their parents' conformity with demographic objectives. Coercive control of bodies is not an admissible instrument of this proposal.

Article 15. Longevity and renewal

Extending healthy life is pursued under conditions of safety and fairness. Lifespan does not confer a right to hold office forever or block access for new generations. Institutions preserve paths for entry, learning, participation, and review. Refusing a medical technology does not diminish a person's status.

Article 16. Research

Society supports research, criticism, replication of results, and open exploration. Restrictions on access or conduct are justified by reasoned, proportionate, and reviewable risks. Economic or political interests alone cannot justify suppressing inconvenient results. People are not turned into involuntary experimental subjects.

Article 17. Future generations

Decisions with long-lasting effects include an assessment of the interests of future generations and civilization's capacity to recover. Their representation is institutionalized without unlimited power over the present. Infrastructure, expertise, and alternatives are preserved. Speculative future benefits do not justify a general suspension of present rights.

Article 18. Exploration and moral status

Exploration is supported within physical limits and responsibilities toward those affected. Expansion does not replace obligations to Earth and does not confer a right to dominate dependent communities. Evidence concerning possible new forms of conscious experience prompts independent examination of the moral and legal protections required.

Article 19. Membership and cooperation

Rules on citizenship, residence, contribution, and access to benefits are public, predictable, and open to challenge. Fundamental rights are not reserved for owners of infrastructure. Institutions cooperate on cross-border effects and the sharing of common burdens without eliminating communities' legitimate autonomy or creating permanently unprotected groups.

Article 20. Verifiable implementation

Every guarantee has an identifiable responsible party, funding, service standards, and a route to remedy. Expansion is based on real capacity and public evaluation. Indicators include autonomy, distribution, continuity, and ecological impact. Savings from automation are calculated after the costs of oversight and remedying harm have been included.

Article 21. Review

The contract is reviewed through legitimate procedures that distinguish principles, implementing laws, and technical rules. Changes that reduce rights or concentrate irreversible powers face stricter scrutiny. No amendment may arbitrarily eliminate equal status or the possibility of future review. The continuity of humanity is not equated with the permanence of any regime.

Conclusion. The conditions of legitimate abundance

This book's proposal is not to trade comfort for submission. It is to exchange cooperation bounded by rights for secure access to the conditions of an autonomous life. People can delegate a very large share of administration without becoming the moral or political

property of administrators. They can choose local, virtual, contemplative, or exploratory lives without constantly proving their economic usefulness.

Such an order has five structural conditions. Production must actually sustain the material promises. Benefits must be distributed through rights and institutions, not only generosity. Power must be observable, contestable, and replaceable. Consumption must respect the ecological conditions of continuity. Rules must preserve future generations' capacity to decide differently. These conditions support one another, but none follows automatically from the others.

Highly productive AI could improve the possibility of meeting them. It would not eliminate conflicts among distribution, freedom, safety, and scarcity. Smarter administration can manage a conflict better; it cannot make all incompatible desires achievable at once. Rigor begins when a promise acknowledges limits and identifies the procedure through which people decide within them.

A very long time horizon also calls for conceptual caution. We cannot guarantee the stability of a civilization for a million years. We can build institutions that reduce certain risks, avoid some dead ends, and preserve more paths forward. The difference between these statements is the difference between a promise impossible to evaluate and a program that can improve through experience.

The project's almost utopian quality does not lie in the disappearance of difficulty. It lies in the possibility that difficulty can, to a much greater degree, be chosen: research without the demand for immediate profit, relationships without coercive material dependency, play without eviction from the shared world, exploration without abandoning those who remain. Neither pleasure, nor creation, nor tranquility should serve as an excuse for the absence of freedom.

The new contract would be justified when a person can receive its benefits while also criticizing its rules, changing providers, challenging its leadership, and proposing another direction. It would be stable when these possibilities remain usable in practice even during crises. It would be worth passing on when the next generation receives not only resources but also the power to manage them according to its own judgments.

This is the proposal's final criterion: abundance becomes legitimate when it reduces dependency, not when it makes dependency more comfortable.

Bibliography and limits of the evidence

The references support specific claims in the text; the selection is not a systematic review of the entire literature. The clauses and analytical models are proposals of this book, not conclusions adopted by the cited institutions. Legal instruments are used as reference points, not as legal advice for a specific situation. Bibliography checked: September 12, 2026.

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[4] Brollo, Fernanda, et al. (2024). Broadening the Gains from Generative AI: The Role of Fiscal Policies. IMF Staff Discussion Note 2024/002.

An analysis of fiscal policy and the distribution of AI benefits. The accounting model and mixed-ownership proposal in this book are distinct developments.

DOI: 10.5089/9798400277177.006

[5] International Energy Agency (2025). Energy and AI. Paris: IEA. Executive summary.

A reference point for energy and infrastructure dependencies. The report's projections are not treated as certainties or as the latest available forecast.

Report and executive summary — IEA

[6] United Nations (1948). Universal Declaration of Human Rights. General Assembly Resolution 217 A (III), December 10, 1948.

A reference point for dignity, equality, and social rights. The declaration's status is not confused with that of a treaty applying identically in every jurisdiction.

Text of the declaration — UN

[7] United Nations (1966). International Covenant on Civil and Political Rights. Adopted by Resolution 2200 A (XXI), December 16, 1966.

A reference point for participation and civil liberties. The proposal concerning delegation of administration is not presented as an exhaustive legal interpretation of the covenant.

Text of the covenant — OHCHR

[8] United Nations (1966). International Covenant on Economic, Social and Cultural Rights. Adopted by Resolution 2200 A (XXI), December 16, 1966.

Relevant to social rights and progressive realization. The book does not treat budget constraints as a general derogation from obligations.

Text of the covenant — OHCHR

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A political-philosophy reference for justifying institutions under fair conditions. The proposed clauses are not derived entirely from Rawls's theory.

Revised edition — Harvard University Press

[10] Ostrom, Elinor (2010). Beyond Markets and States: Polycentric Governance of Complex Economic Systems. American Economic Review, 100(3), 641–672.

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[12] Ministry of Social Affairs and Health, Finland (2020). Results of the basic income experiment: small employment effects, better perceived economic security and mental wellbeing. Press release, May 6, 2020.

Official summary of the Finnish experiment. It does not establish the feasibility of a sufficient universal income in an economy without wages.

Results of the experiment — Finnish ministry

[13] Norges Bank Investment Management. About the fund. Official presentation of the Government Pension Fund Global. Accessed September 12, 2026.

A precedent for publicly owned capital managed over the long term. The spending rule and expected return are not a guarantee of returns.

Fund presentation — NBIM

[14] European Union (2022). Regulation (EU) 2022/2065 on a Single Market for Digital Services (Digital Services Act). In particular, Articles 25 and 38.

Limited reference points concerning interface design and certain recommender systems. The obligations do not apply identically to all services.

Text of the regulation — EUR-Lex

[15] European Union (2016). Regulation (EU) 2016/679 on the protection of personal data (GDPR). In particular, Articles 5 and 25.

A reference point for purpose limitation, data minimization, and protection by design. Rights and exceptions require contextual analysis.

Text of the regulation — EUR-Lex

[16] European Union (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI Act). Article 86; consolidated text consulted, dated July 27, 2026.

The right mentioned concerns decisions and systems specified by the regulation, subject to its limits. The consolidated text is documentary; authentic acts and the application timetable must be checked for legal use.

Consolidated text — EUR-Lex

[17] National Institute of Standards and Technology (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1.

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DOI: 10.6028/NIST.AI.100-1

[18] Richardson, Katherine, et al. (2023). Earth beyond six of nine planetary boundaries. Science Advances, 9(37), eadh2458.

The 2023 assessment supports a multidimensional approach to ecological risks; by itself, it does not establish a political distribution of budgets.

DOI: 10.1126/sciadv.adh2458

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A reference point for cumulative effects and irreversible changes. It is not used for institutional forecasts over millions of years.

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[20] United Nations, Department of Economic and Social Affairs, Population Division (2024). World Population Prospects 2024: Summary of Results. UN DESA/POP/2024/TR/NO. 9.

The projections are conditional and have a limited time horizon. The relationship between births, deaths, and population is analyzed separately from the statistical scenarios.

World Population Prospects 2024 — UN

[21] UNFPA (2025). State of World Population 2025: The Real Fertility Crisis — The Pursuit of Reproductive Agency in a Changing World.

A reference point for reproductive autonomy and barriers to fulfilling intentions. It does not, by itself, resolve the tensions of a radical-longevity scenario.

2025 report — UNFPA

[22] United Nations General Assembly (2024). Pact for the Future. Resolution A/RES/79/1, September 22, 2024. Annex II: Declaration on Future Generations.

An international political commitment. It is not presented as a global constitution or an independent enforcement mechanism.

Declaration on Future Generations — UN

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Outer Space Treaty — UNOOSA

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